

APPLIED MATH 3

MISC

INTRO TO UNREAL CLASSES FOR MOVEMENT

Review & Basics – Input and Movement in Unreal

ASIDE: DELTA TIME

- The idea of Delta Time can be confusing at first, however it is important to understand as it has many uses.
- The Delta time is the length of time it took to render the previous frame.
- It is also the inverse of the frame rate.
 - For example, if you have a frame rate of 60 frame per second (FPS), you would have a Delta time of $1/60$ of a second (on average)

ASIDE: DELTA TIME

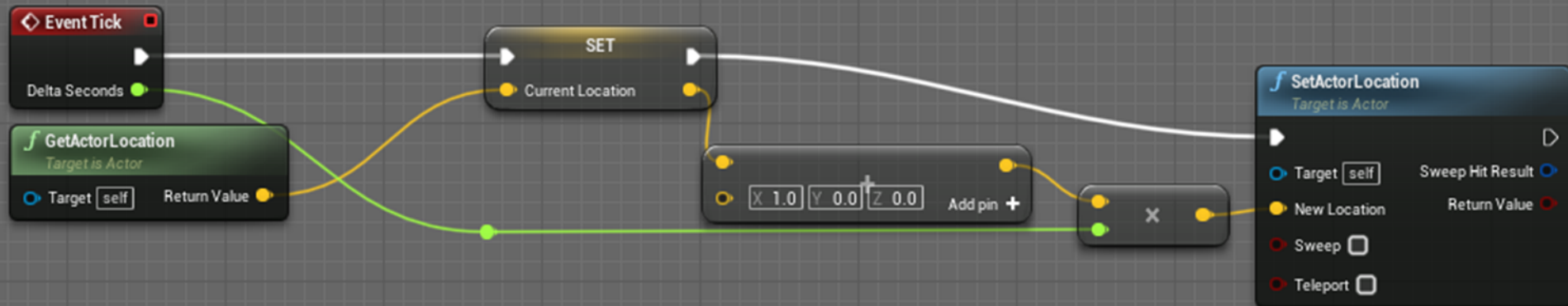
- You can use this delta time value to normalize the movement speed, even if the frame rate is changing.
- For example, in our previous movement example, our Pawn would move 1 unit per frame in the X axis. This means if the framerate was 30 fps, we would move 30 units per second. If the frame rate was 100 fps, it would move 100 units per second.
- This is also compounded with the fact that the frame rate fluctuates over time, making the movement inconsistent.

ASIDE: DELTA TIME

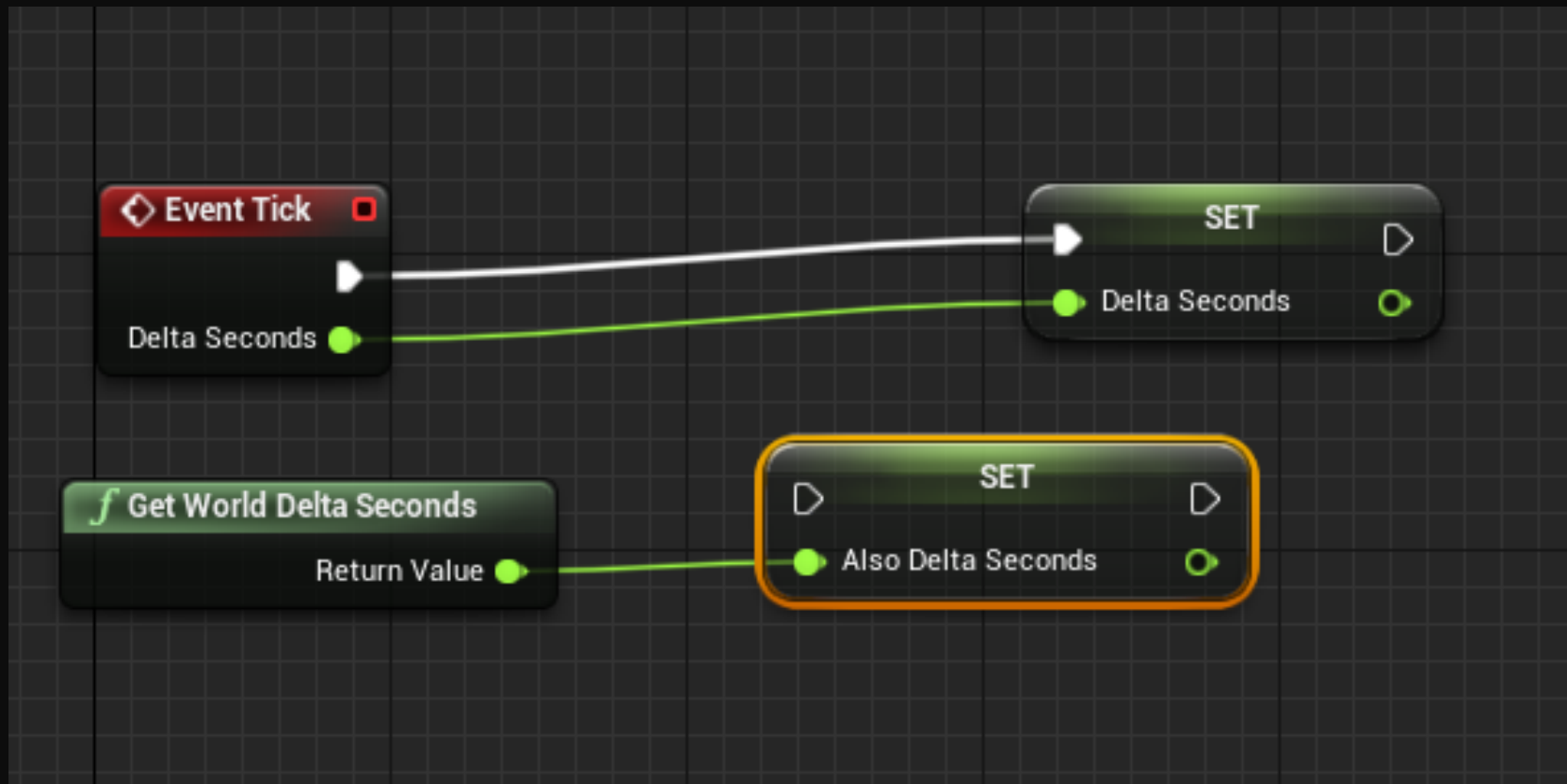
- The Simplest way of thinking of delta time is that it is the inverse of the frame rate.
 - $60 \text{ FPS} = 1/60 \text{ deltatime}$
- Commonly it is used to convert a unit per frame into a unit per second.
- For example, if our vector3 position is being updated by adding our speed variable (1,0,0) with a frame rate of 60 fps, it would move 60 units in 1 second.
- If we multiply our speed variable by the delta time, it would instead move 1 unit per second, regardless of the frame rate.
 - $\text{Speed} * \text{deltatime} = \text{speed per second}$

ASIDE: DELTA TIME

Notes: This Pawn will move 1 Unit per Second in the X Axis



DELTA TIME IN BLUEPRINTS



ASIDE: SQUARED VECTOR LENGTH

- Aside: Squared Magnitude
- You will sometimes see the use of Squared Magnitude when working in games.
- Typically this is used as an optimization to avoid square root operations when they are not necessary, as calculating a square root is an expensive operation.
- When finding the length of a vector (as we saw), the final operation is a square root, so if we are simply comparing the values to another vector to see which is longer, the squared version can be used instead.

